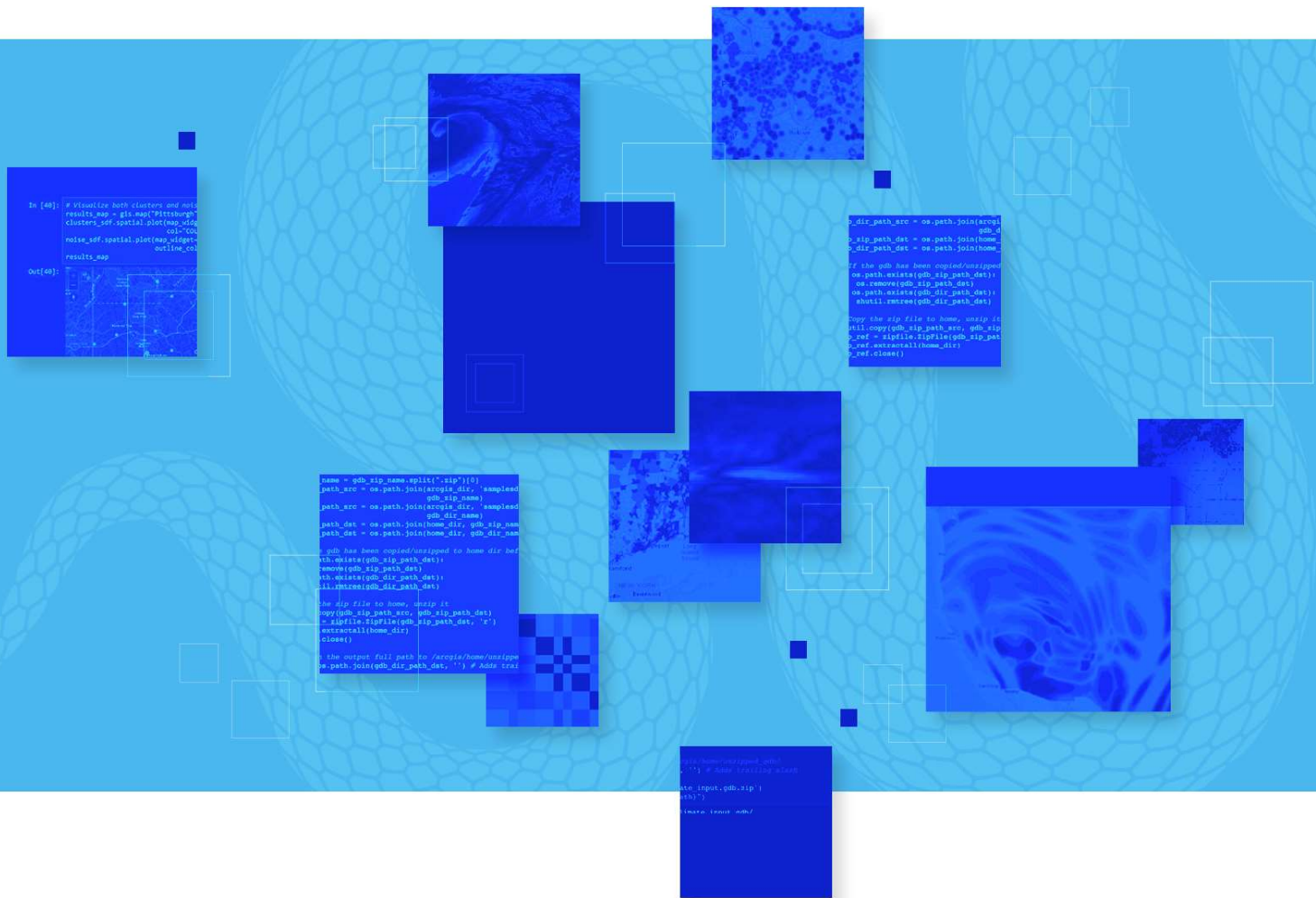




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9.0

# ArcGIS® Notebooks

## Python Libraries Manifest

ArcGIS® Notebooks provides a spatially optimized Jupyter experience to create, share, and run data science, data management, and administrative scripts.

With ArcGIS Notebooks, you can perform analysis, automate workflows, and immediately visualize data and analysis results in a geographic context. It is an efficient, modern environment that combines code, real-time visualizations and maps, and data tools.

ArcGIS Notebooks features the ArcGIS Python libraries, the ArcGIS API for Python, and ArcPy™. ArcGIS Notebooks also includes access to hundreds of open-source Python libraries and packages.

This publication provides a complete list of the Python libraries—including their versions—that are packaged in the ArcGIS Notebooks runtime version 9.0. The first two lists are for Linux Containers, the second two are for Windows Containers.

# ArcGIS Notebooks Runtime

Python 3 Standard (Linux Containers) | Version 9.0

## RELEASED

October 2023

## TOTAL PACKAGES

538

## SUPPORTED IN

ArcGIS Enterprise 11.2  
and ArcGIS Online  
October 2023.R03 release

In ArcGIS Enterprise, you can  
customize your notebook  
environment, learn more at  
[Extend Notebook Runtime](#).

| Library Name         | Version    | Library Name            | Version   | Library Name              | Version  |
|----------------------|------------|-------------------------|-----------|---------------------------|----------|
| _libgcc_mutex        | 0.1        | cffi                    | 1.15.1    | etuples                   | 0.3.9    |
| _openmp_mutex        | 5.1        | cfitsio                 | 3.470     | exceptiongroup            | 1.0.4    |
| _py-xgboost-mutex    | 2.0        | cftime                  | 1.6.2     | executing                 | 0.8.3    |
| _tflow_select        | 2.7.0      | charls                  | 2.2.0     | expat                     | 2.5.0    |
| abseil-cpp           | 20211102.0 | charset-normalizer      | 2.0.4     | fairlearn                 | 0.8.0    |
| absl-py              | 1.4.0      | click                   | 8.0.4     | fastai                    | 1.0.63   |
| addict               | 2.4.0      | click-plugins           | 1.1.1     | fastprogress              | 0.2.3    |
| affine               | 2.3.0      | cliff                   | 3.8.0     | fasttext                  | 0.9.2    |
| aiohttp              | 3.8.5      | cligj                   | 0.7.2     | ffmpeg                    | 4.0      |
| aiosignal            | 1.2.0      | cloudpickle             | 2.2.1     | filelock                  | 3.9.0    |
| albumations          | 1.0.3      | cmaes                   | 0.8.2     | fiona                     | 1.8.21   |
| alembic              | 1.8.1      | cmd2                    | 2.1.1     | fire                      | 0.4.0    |
| altair               | 4.1.0      | colorama                | 0.4.6     | fmt                       | 9.1.0    |
| ansiwrap             | 0.8.4      | colorcet                | 3.0.1     | font-ttf-dejavu-sans-mono | 2.37     |
| anyio                | 3.5.0      | colorlog                | 5.0.1     | font-ttf-inconsolata      | 2.001    |
| aom                  | 3.6.0      | colour                  | 0.1.5     | font-ttf-source-code-pro  | 2.030    |
| arcgis               | 2.2.0.1    | comm                    | 0.1.2     | font-ttf-ubuntu           | 0.83     |
| argon2-cffi          | 21.3.0     | conda                   | 23.7.2    | fontconfig                | 2.14.1   |
| argon2-cffi-bindings | 21.2.0     | conda-content-trust     | 0.2.0     | fonts-anaconda            | 1        |
| arviz                | 0.16.0     | conda-libmamba-solver   | 23.7.0    | fonts-conda-ecosystem     | 1        |
| astor                | 0.8.1      | conda-package-handling  | 2.2.0     | fonttools                 | 4.25.0   |
| asttokens            | 2.0.5      | conda-package-streaming | 0.9.0     | freelut                   | 3.0.0    |
| astunparse           | 1.6.3      | cons                    | 0.4.6     | freetype                  | 2.12.1   |
| async-timeout        | 4.0.2      | contourpy               | 1.0.5     | freexl                    | 1.0.6    |
| atk-1.0              | 2.36.0     | cryptography            | 41.0.3    | fribidi                   | 1.0.10   |
| attrs                | 23.1.0     | cuda-toolkit            | 11.1.1    | frozenlist                | 1.3.3    |
| azure-core           | 1.12.0     | cudnn                   | 8.1.0.77  | fsspec                    | 2023.9.2 |
| azure-storage-blob   | 12.8.0     | curl                    | 8.2.1     | gast                      | 0.5.3    |
| backcall             | 0.2.0      | cycler                  | 0.11.0    | gdal                      | 3.4.1    |
| beautifulsoup4       | 4.12.2     | cymem                   | 2.0.6     | gdk-pixbuf                | 2.42.10  |
| black                | 23.3.0     | cython                  | 0.29.33   | geomet                    | 1.0.0    |
| blas                 | 1.0        | cython-blis             | 0.4.1     | geos                      | 3.8.0    |
| bleach               | 4.1.0      | cytoolz                 | 0.12.0    | geotiff                   | 1.7.0    |
| blinker              | 1.4        | dask                    | 2023.3.2  | giflib                    | 5.2.1    |
| blosc                | 1.21.3     | dask-core               | 2023.3.2  | glib                      | 2.69.1   |
| bokeh                | 2.4.3      | dask-glm                | 0.2.0     | gmp                       | 6.2.1    |
| boltons              | 23.0.0     | dask-ml                 | 2023.3.24 | gmpy2                     | 2.1.2    |
| boost                | 1.79.0     | dataclasses             | 0.8       | gobject-introspection     | 1.72.0   |
| boto3                | 1.24.28    | dav1d                   | 1.2.1     | google-auth               | 2.22.0   |
| botocore             | 1.27.59    | dbus                    | 1.13.18   | google-auth-oauthlib      | 0.4.1    |
| bottleneck           | 1.3.5      | debugpy                 | 1.6.7     | google-pasta              | 0.2.0    |
| brtli                | 1.0.9      | decorator               | 5.1.1     | googledrivedownloader     | 0.4      |
| brtli-bin            | 1.0.9      | defusedxml              | 0.7.1     | graphite2                 | 1.3.14   |
| brtli-py             | 0.7.0      | descartes               | 1.1.0     | graphviz                  | 2.50.0   |
| brunli               | 0.1        | dill                    | 0.3.6     | greenlet                  | 2.0.1    |
| bzip2                | 1.0.8      | distributed             | 2023.3.2  | grpc-cpp                  | 1.48.2   |
| c-ares               | 1.19.1     | dm-tree                 | 0.1.7     | grpcio                    | 1.48.2   |
| ca-certificates      | 2023.08.22 | dtreeviz                | 1.3.7     | gtk2                      | 2.24.33  |
| cachetools           | 4.2.2      | einops                  | 0.3.2     | gts                       | 0.7.6    |
| cairo                | 1.16.0     | ensemble-boxes          | 1.0.8     | h3-py                     | 3.7.3    |
| catalogue            | 1.0.0      | entrypoints             | 0.4       | h5netcdf                  | 1.2.0    |
| catboost             | 0.26       | et_xmlfile              | 1.1.0     | h5py                      | 3.7.0    |
| category_encoders    | 2.2.2      |                         |           |                           |          |
| certifi              | 2023.7.22  |                         |           |                           |          |

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| Library Name                      | Version   | Library Name     | Version      | Library Name                     | Version  |
|-----------------------------------|-----------|------------------|--------------|----------------------------------|----------|
| harfbuzz                          | 4.3.0     | krb5             | 1.20.1       | libzopfli                        | 1.0.3    |
| hdf4                              | 4.2.13    | laspy            | 1.7.0        | lightgbm                         | 3.3.5    |
| hdf5                              | 1.10.6    | lcms2            | 2.12         | llvmlite                         | 0.40.0   |
| holoviews                         | 1.17.1    | ld_impl_linux-64 | 2.38         | locket                           | 1.0.0    |
| html5lib                          | 1.1       | lerc             | 3.0          | logical-unification              | 0.4.6    |
| huggingface_hub                   | 0.10.1    | libaec           | 1.0.4        | lxml                             | 4.9.3    |
| icu                               | 58.2      | libarchive       | 3.6.2        | lz4                              | 4.3.2    |
| idna                              | 3.4       | libavif          | 0.11.1       | lz4-c                            | 1.9.4    |
| imagecodecs                       | 2023.1.23 | libboost         | 1.79.0       | lzo                              | 2.10     |
| imageio                           | 2.31.4    | libbrotlicommon  | 1.0.9        | mako                             | 1.2.3    |
| imgaug                            | 0.4.0     | libbrotlidec     | 1.0.9        | markdown                         | 3.4.1    |
| importlib-metadata                | 6.0.0     | libbrotlienc     | 1.0.9        | markdown-it-py                   | 2.2.0    |
| importlib_metadata                | 6.0.0     | libcurl          | 8.2.1        | markupsafe                       | 2.1.1    |
| iniconfig                         | 1.1.1     | libdap4          | 3.19.1       | matplotlib                       | 3.6.0    |
| inplace-abn                       | 1.1.0     | libdeflate       | 1.17         | matplotlib-base                  | 3.6.0    |
| intel-openmp                      | 2023.1.0  | libedit          | 3.1.20221030 | matplotlib-inline                | 0.1.6    |
| ipykernel                         | 6.19.2    | libev            | 4.33         | mdurl                            | 0.1.0    |
| ipympl                            | 0.8.7     | libffi           | 3.4.4        | minikanren                       | 1.0.3    |
| ipython                           | 7.34.0    | libgcc-ng        | 11.2.0       | mistune                          | 2.0.4    |
| ipython_genutils                  | 0.2.0     | libgd            | 2.3.3        | mkl                              | 2023.1.0 |
| ipywidgets                        | 7.7.2     | libgdal          | 3.4.1        | mkl-service                      | 2.4.0    |
| isodate                           | 0.6.0     | libgfortran-ng   | 11.2.0       | mkl_fft                          | 1.3.8    |
| jaraco.classes                    | 3.2.1     | libgfortran5     | 11.2.0       | mkl_random                       | 1.2.4    |
| jasper                            | 2.0.14    | libglu           | 9.0.0        | mljar-supervised                 | 0.11.2   |
| jedi                              | 0.18.2    | libgomp          | 11.2.0       | mmcls                            | 0.25.0   |
| jeepney                           | 0.7.1     | libllvm14        | 14.0.6       | mmcv-full                        | 1.6.0    |
| jinja2                            | 3.1.2     | libmamba         | 1.5.1        | mmdet                            | 2.27.0   |
| jmespath                          | 0.10.0    | libmambapy       | 1.5.1        | mmdet3d                          | 1.0.0rc5 |
| joblib                            | 1.2.0     | libnetcdf        | 4.8.1        | mmengine                         | 0.7.3    |
| jpeg                              | 9e        | libnghttp2       | 1.52.0       | mmsegmentation                   | 0.30.0   |
| json-c                            | 0.16      | libopencv        | 4.5.2        | mock                             | 4.0.3    |
| jsonpatch                         | 1.32      | libopus          | 1.3.1        | more-itertools                   | 8.12.0   |
| jsonpointer                       | 2.1       | libpng           | 1.6.39       | motmetrics                       | 1.1.3    |
| jsonschema                        | 4.17.3    | libpq            | 12.15        | mpc                              | 1.1.0    |
| jupyter_client                    | 6.1.12    | libprotobuf      | 3.20.3       | mpfr                             | 4.0.2    |
| jupyter_contrib_core              | 0.4.0     | librsvg          | 2.54.4       | mpmath                           | 1.3.0    |
| jupyter_contrib_nbextensions      | 0.5.1     | libsodium        | 1.0.18       | msgpack-python                   | 1.0.3    |
| jupyter_core                      | 4.11.2    | libsolv          | 0.7.24       | msrest                           | 0.6.21   |
| jupyter_highlight_selected_word   | 0.2.0     | libspatialindex  | 1.9.3        | multi-scale-deformable-attention | 1.0.0    |
| jupyter_latex_envs                | 1.4.6     | libspatialite    | 4.3.0a       | multidict                        | 6.0.2    |
| jupyter_nbextensions_configurator | 0.4.1     | libssh2          | 1.10.0       | multipledispatch                 | 0.6.0    |
| jupyter_server                    | 1.21.0    | libstdcxx-ng     | 11.2.0       | munch                            | 2.5.0    |
| jupyterlab_pygments               | 0.2.2     | libtiff          | 4.5.1        | munkres                          | 1.1.4    |
| jupyterlab_widgets                | 3.0.3     | libtool          | 2.4.6        | murmurhash                       | 1.0.7    |
| jxrlib                            | 1.1       | libuuid          | 1.41.5       | mypy_extensions                  | 1.0.0    |
| kealib                            | 1.5.0     | libuv            | 1.44.2       | nb_conda_kernels                 | 2.3.1    |
| keepalive                         | 0.5       | libvpx           | 1.7.0        | nbclassic                        | 0.5.5    |
| keras-base                        | 2.7.0     | libwebp          | 1.3.2        | nbclient                         | 0.7.0    |
| keras-gpu                         | 2.7.0     | libwebp-base     | 1.3.2        | nbconvert                        | 7.2.1    |
| keras-preprocessing               | 1.1.2     | libxcb           | 1.15         | nbformat                         | 5.6.1    |
| keyring                           | 23.13.1   | libxgboost       | 1.5.0        | ncurses                          | 6.4      |
| kiwisolver                        | 1.4.4     | libxml2          | 2.10.4       | nest-asyncio                     | 1.5.6    |
|                                   |           | libxslt          | 1.1.37       | netcdf4                          | 1.6.2    |
|                                   |           | libzip           | 1.8.0        |                                  |          |

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| Library Name               | Version | Library Name          | Version      | Library Name            | Version  |
|----------------------------|---------|-----------------------|--------------|-------------------------|----------|
| networkx                   | 3.1     | ptyprocess            | 0.7.0        | requests-gssapi         | 1.2.3    |
| neural-structured-learning | 1.4.0   | pure_eval             | 0.2.2        | requests-kerberos       | 0.12.0   |
| ninja                      | 1.10.2  | py-boost              | 1.79.0       | requests-oauthlib       | 1.3.0    |
| ninja-base                 | 1.10.2  | py-opencv             | 4.5.2        | requests-toolbelt       | 1.0.0    |
| notebook                   | 6.5.4   | py-xgboost            | 1.5.0        | rich                    | 13.3.5   |
| notebook-shim              | 0.1.0   | pyasn1                | 0.4.8        | rise                    | 5.7.1    |
| numba                      | 0.57.1  | pyasn1-modules        | 0.2.8        | rsa                     | 4.7.2    |
| numexpr                    | 2.8.4   | pybind11              | 2.10.4       | rtree                   | 0.9.4    |
| numpy                      | 1.22.3  | pybind11-abi          | 4            | ruamel.yaml             | 0.17.21  |
| numpy-base                 | 1.22.3  | pybind11-global       | 2.10.4       | ruamel.yaml.clib        | 0.2.6    |
| nuscenes-devkit            | 1.1.3   | pycocotools           | 2.0.2        | s3transfer              | 0.6.0    |
| nvidia-ml-py3              | 7.352.0 | pycosat               | 0.6.4        | sacremoses              | 0.0.43   |
| oauthlib                   | 3.2.2   | pycparser             | 2.21         | scikit-image            | 0.19.3   |
| onnx                       | 1.13.0  | pyct                  | 0.5.0        | scikit-learn            | 1.2.2    |
| onnx-tf                    | 1.8.0   | pygments              | 2.14.0       | scikit-plot             | 0.3.7    |
| opencv                     | 4.5.2   | pyjwt                 | 2.4.0        | scipy                   | 1.9.3    |
| openjpeg                   | 2.4.0   | pykerberos            | 1.2.1        | seaborn                 | 0.12.2   |
| openpyxl                   | 3.0.10  | pylerc                | 3.0          | secretstorage           | 3.3.1    |
| openssl                    | 3.0.11  | pymc                  | 5.6.1        | send2trash              | 1.8.0    |
| opt_einsum                 | 3.3.0   | pyodbc                | 4.0.34       | sentencepiece           | 0.1.99   |
| optuna                     | 2.8.0   | pyopenssl             | 23.2.0       | setuptools              | 68.0.0   |
| packaging                  | 23.1    | pyparsing             | 3.0.9        | shap                    | 0.39.0   |
| pandas                     | 2.0.2   | pyperclip             | 1.8.2        | shapely                 | 2.0.1    |
| pandocfilters              | 1.5.0   | pyquaternion          | 0.9.9        | simplejson              | 3.17.6   |
| panel                      | 0.14.3  | pyrsistent            | 0.18.0       | six                     | 1.16.0   |
| pango                      | 1.50.7  | pyshp                 | 2.1.3        | slicer                  | 0.0.7    |
| papermill                  | 2.3.4   | pysocks               | 1.7.1        | snappy                  | 1.1.9    |
| param                      | 1.13.0  | pyspnego              | 0.8.0        | sniffio                 | 1.2.0    |
| parso                      | 0.8.3   | pytables              | 3.7.0        | snuggs                  | 1.4.7    |
| partd                      | 1.4.0   | pytensor              | 2.12.3       | sortedcontainers        | 2.4.0    |
| pathspec                   | 0.9.0   | pytest                | 7.4.0        | soupsieve               | 2.4      |
| patsy                      | 0.5.3   | python                | 3.9.18       | spacy                   | 2.2.4    |
| pbr                        | 5.6.0   | python-dateutil       | 2.8.2        | sparqlwrapper           | 1.8.5    |
| pcre                       | 8.45    | python-fastjsonschema | 2.16.2       | sqlalchemy              | 1.4.39   |
| pcre2                      | 10.42   | python-flatbuffers    | 1.12         | sqlite                  | 3.41.2   |
| pexpect                    | 4.8.0   | python-graphviz       | 0.20.1       | srsly                   | 1.0.5    |
| pickleshare                | 0.7.5   | python-gssapi         | 1.8.1        | stack_data              | 0.5.1    |
| pillow                     | 9.4.0   | python-sounddevice    | 0.4.4        | statsmodels             | 0.13.5   |
| pip                        | 23.2.1  | python_abi            | 3.9          | stevedore               | 3.3.0    |
| pixman                     | 0.40.0  | pytorch               | 1.8.2        | sympy                   | 1.11.1   |
| plac                       | 1.1.0   | pytz                  | 2023.3.post1 | tabulate                | 0.8.9    |
| platformdirs               | 3.10.0  | pyviz_comms           | 2.3.0        | tbb                     | 2021.8.0 |
| plotly                     | 5.15.0  | pywavelets            | 1.4.1        | tblib                   | 1.7.0    |
| pluggy                     | 1.0.0   | pyyaml                | 6.0          | tenacity                | 8.2.2    |
| portaudio                  | 19.6.0  | pyzmq                 | 24.0.1       | tensorboard             | 2.7.0    |
| postgresql                 | 12.15   | rasterio              | 1.2.10       | tensorboard-data-server | 0.6.1    |
| preshed                    | 3.0.6   | rasterstats           | 0.16.0       | tensorboard-plugin-wit  | 1.8.1    |
| prettytable                | 2.1.0   | rdflib                | 5.0.0        | tensorboardx            | 2.2      |
| proj                       | 6.2.1   | re2                   | 2022.04.01   | tensorflow-addons       | 0.15.0   |
| prometheus_client          | 0.8.0   | readline              | 8.2          | tensorflow-estimator    | 2.7.0    |
| prompt_toolkit             | 3.0.5   | regex                 | 2022.7.9     | tensorflow-gpu          | 2.7.0    |
| protobuf                   | 3.20.3  | reproc                | 14.2.4       | tensorflow-hub          | 0.12.0   |
| psutil                     | 5.9.0   | reproc-cpp            | 14.2.4       |                         |          |
|                            |         | requests              | 2.31.0       |                         |          |



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| tensorflow-model-optimization | 0.7.3     | yapf         | 0.31.0  |
| termcolor                     | 2.1.0     | yaml         | 1.8.1   |
| terminado                     | 0.17.1    | zeromq       | 4.3.4   |
| terminaltables                | 3.1.0     | zfp          | 1.0.0   |
| textwrap3                     | 0.9.2     | zict         | 3.0.0   |
| tfllite-model-maker           | 0.3.4     | zipp         | 3.11.0  |
| tfllite-support               | 0.4.1     | zlib         | 1.2.13  |
| thinc                         | 7.4.0     | zstandard    | 0.19.0  |
| threadpoolctl                 | 2.2.0     | zstd         | 1.5.5   |
| tifffile                      | 2023.4.12 |              |         |
| tiledb                        | 2.3.3     |              |         |
| timm                          | 0.4.12    |              |         |
| tinycss2                      | 1.1.1     |              |         |
| tk                            | 8.6.12    |              |         |
| tokenizers                    | 0.10.1    |              |         |
| tomli                         | 2.0.1     |              |         |
| toolz                         | 0.12.0    |              |         |
| torch-cluster                 | 1.5.9     |              |         |
| torch-geometric               | 1.7.2     |              |         |
| torch-scatter                 | 2.0.7     |              |         |
| torch-sparse                  | 0.6.10    |              |         |
| torch-spline-conv             | 1.2.1     |              |         |
| torchvision                   | 0.9.2     |              |         |
| tornado                       | 6.3.2     |              |         |
| tqdm                          | 4.65.0    |              |         |
| traitlets                     | 5.5.0     |              |         |
| transformers                  | 4.14.1    |              |         |
| trimesh                       | 2.35.39   |              |         |
| typeguard                     | 2.13.0    |              |         |
| typing-extensions             | 4.7.1     |              |         |
| typing_extensions             | 4.7.1     |              |         |
| tzdata                        | 2023c     |              |         |
| ujson                         | 5.4.0     |              |         |
| unixodbc                      | 2.3.11    |              |         |
| urllib3                       | 1.26.16   |              |         |
| wasabi                        | 0.9.1     |              |         |
| wcwidth                       | 0.2.5     |              |         |
| webencodings                  | 0.5.1     |              |         |
| websocket-client              | 0.58.0    |              |         |
| werkzeug                      | 2.2.3     |              |         |
| wheel                         | 0.41.2    |              |         |
| widetsnbextension             | 3.6.1     |              |         |
| wordcloud                     | 1.8.1     |              |         |
| wrapt                         | 1.14.1    |              |         |
| xarray                        | 2023.6.0  |              |         |
| xarray-einstats               | 0.6.0     |              |         |
| xerces-c                      | 3.2.4     |              |         |
| xgboost                       | 1.5.0     |              |         |
| xlrd                          | 2.0.1     |              |         |
| xmltodict                     | 0.12.0    |              |         |
| xz                            | 5.4.2     |              |         |
| yaml                          | 0.2.5     |              |         |
| yaml-cpp                      | 0.7.0     |              |         |

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# ArcGIS Notebooks Runtime

Python 3 Advanced (Linux Containers) | Version 9.0

## RELEASED

October 2023

## TOTAL PACKAGES

544

## SUPPORTED IN

ArcGIS Enterprise 11.2 and ArcGIS Online October 2023.R03 release.

The use of the Advanced and/or Advanced with GPU runtime consumes credits within ArcGIS Online. See ArcGIS Online [credits by capability](#) for more details.

In ArcGIS Enterprise you can customize your notebook environment, learn more at [Extend Notebook Runtime](#).

| Library Name         | Version    | Library Name            | Version   | Library Name              | Version  |
|----------------------|------------|-------------------------|-----------|---------------------------|----------|
| _libgcc_mutex        | 0.1        | cffi                    | 1.15.1    | exceptiongroup            | 1.0.4    |
| _openmp_mutex        | 5.1        | cfitsio                 | 3.470     | executing                 | 0.8.3    |
| _py-xgboost-mutex    | 2.0        | cftime                  | 1.6.2     | expat                     | 2.5.0    |
| _tflow_select        | 2.7.0      | charls                  | 2.2.0     | fairlearn                 | 0.8.0    |
| abseil-cpp           | 20211102.0 | charset-normalizer      | 2.0.4     | fastai                    | 1.0.63   |
| absl-py              | 1.4.0      | click                   | 8.0.4     | fastprogress              | 0.2.3    |
| addict               | 2.4.0      | click-plugins           | 1.1.1     | fasttext                  | 0.9.2    |
| affine               | 2.3.0      | cliff                   | 3.8.0     | ffmpeg                    | 4.0      |
| aiohttp              | 3.8.5      | cligj                   | 0.7.2     | filelock                  | 3.9.0    |
| aiosignal            | 1.2.0      | cloudpickle             | 2.2.1     | fiona                     | 1.8.21   |
| albumations          | 1.0.3      | cmaes                   | 0.8.2     | fire                      | 0.4.0    |
| alembic              | 1.8.1      | cmd2                    | 2.1.1     | fmt                       | 9.1.0    |
| altair               | 4.1.0      | colorama                | 0.4.6     | font-ttf-dejavu-sans-mono | 2.37     |
| ansiwrap             | 0.8.4      | colorcet                | 3.0.1     | font-ttf-inconsolata      | 2.001    |
| anyio                | 3.5.0      | colorlog                | 5.0.1     | font-ttf-source-code-pro  | 2.030    |
| aom                  | 3.6.0      | colour                  | 0.1.5     | font-ttf-ubuntu           | 0.83     |
| arcgis               | 2.2.0.1    | comm                    | 0.1.2     | fontconfig                | 2.14.1   |
| arcgis-server-py3    | 11.2.0     | conda                   | 23.7.2    | fonts-anaconda            | 1        |
| argon2-cffi          | 21.3.0     | conda-content-trust     | 0.2.0     | fonts-conda-ecosystem     | 1        |
| argon2-cffi-bindings | 21.2.0     | conda-libmamba-solver   | 23.7.0    | fonttools                 | 4.25.0   |
| arviz                | 0.16.0     | conda-package-handling  | 2.2.0     | freeglut                  | 3.0.0    |
| astor                | 0.8.1      | conda-package-streaming | 0.9.0     | freetype                  | 2.12.1   |
| asttokens            | 2.0.5      | cons                    | 0.4.6     | freexl                    | 1.0.6    |
| astunparse           | 1.6.3      | contourpy               | 1.0.5     | fribidi                   | 1.0.10   |
| async-timeout        | 4.0.2      | cryptography            | 41.0.3    | frozenlist                | 1.3.3    |
| atk-1.0              | 2.36.0     | cuda-toolkit            | 11.1.1    | fsspec                    | 2023.9.2 |
| attrs                | 23.1.0     | cuda-nn                 | 8.1.0.77  | future                    | 0.18.3   |
| azure-core           | 1.12.0     | curl                    | 8.2.1     | gast                      | 0.5.3    |
| azure-storage-blob   | 12.8.0     | cycler                  | 0.11.0    | gdal                      | 3.4.1    |
| backcall             | 0.2.0      | cymem                   | 2.0.6     | gdk-pixbuf                | 2.42.10  |
| beautifulsoup4       | 4.12.2     | cython                  | 0.29.33   | geomet                    | 1.0.0    |
| black                | 23.3.0     | cython-blis             | 0.4.1     | geos                      | 3.8.0    |
| blas                 | 1.0        | cytoolz                 | 0.12.0    | geotiff                   | 1.7.0    |
| bleach               | 4.1.0      | dask                    | 2023.3.2  | giflib                    | 5.2.1    |
| blinker              | 1.4        | dask-core               | 2023.3.2  | glib                      | 2.69.1   |
| blosc                | 1.21.3     | dask-glm                | 0.2.0     | gmp                       | 6.2.1    |
| bokeh                | 2.4.3      | dask-ml                 | 2023.3.24 | gmpy2                     | 2.1.2    |
| boltons              | 23.0.0     | dataclasses             | 0.8       | gobject-introspection     | 1.72.0   |
| boost                | 1.79.0     | dav1d                   | 1.2.1     | google-auth               | 2.22.0   |
| boto3                | 1.24.28    | dbus                    | 1.13.18   | google-auth-oauthlib      | 0.4.1    |
| botocore             | 1.27.59    | debugpy                 | 1.6.7     | google-pasta              | 0.2.0    |
| bottleneck           | 1.3.5      | decorator               | 5.1.1     | googledrivedownloader     | 0.4      |
| brtli                | 1.0.9      | defusedxml              | 0.7.1     | graphite2                 | 1.3.14   |
| brtli-bin            | 1.0.9      | descartes               | 1.1.0     | graphviz                  | 2.50.0   |
| brtli-py             | 0.7.0      | dill                    | 0.3.6     | greenlet                  | 2.0.1    |
| brunsli              | 0.1        | distributed             | 2023.3.2  | grpc-cpp                  | 1.48.2   |
| bzip2                | 1.0.8      | dm-tree                 | 0.1.7     | grpcio                    | 1.48.2   |
| c-ares               | 1.19.1     | dtreeviz                | 1.3.7     | gtk2                      | 2.24.33  |
| ca-certificates      | 2023.08.22 | einops                  | 0.3.2     | gts                       | 0.7.6    |
| cachetools           | 4.2.2      | ensemble-boxes          | 1.0.8     | h3-py                     | 3.7.3    |
| cairo                | 1.16.0     | entrypoints             | 0.4       | h5netcdf                  | 1.2.0    |
| catalogue            | 1.0.0      | et-xmlfile              | 1.1.0     | h5py                      | 3.7.0    |
| catboost             | 0.26       | etuples                 | 0.3.9     | harfbuzz                  | 4.3.0    |
| category-encoders    | 2.2.2      |                         |           |                           |          |
| certifi              | 2023.7.22  |                         |           |                           |          |

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| Library Name                      | Version   | Library Name     | Version      | Library Name                     | Version  |
|-----------------------------------|-----------|------------------|--------------|----------------------------------|----------|
| hdf4                              | 4.2.13    | laspy            | 1.7.0        | llvmlite                         | 0.40.0   |
| hdf5                              | 1.10.6    | lcms2            | 2.12         | locket                           | 1.0.0    |
| holoviews                         | 1.17.1    | ld_impl_linux-64 | 2.38         | logical-unification              | 0.4.6    |
| html5lib                          | 1.1       | lerc             | 3.0          | lxml                             | 4.9.3    |
| huggingface_hub                   | 0.10.1    | libaec           | 1.0.4        | lz4                              | 4.3.2    |
| icu                               | 58.2      | libarchive       | 3.6.2        | lz4-c                            | 1.9.4    |
| idna                              | 3.4       | libavif          | 0.11.1       | lzo                              | 2.10     |
| imagecodecs                       | 2023.1.23 | libboost         | 1.79.0       | mako                             | 1.2.3    |
| imageio                           | 2.31.4    | libbrotlicommon  | 1.0.9        | markdown                         | 3.4.1    |
| imgaug                            | 0.4.0     | libbrotlidec     | 1.0.9        | markdown-it-py                   | 2.2.0    |
| importlib-metadata                | 6.0.0     | libbrotlienc     | 1.0.9        | markupsafe                       | 2.1.1    |
| importlib_metadata                | 6.0.0     | libcurl          | 8.2.1        | matplotlib                       | 3.6.0    |
| iniconfig                         | 1.1.1     | libdap4          | 3.19.1       | matplotlib-base                  | 3.6.0    |
| inplace-abn                       | 1.1.0     | libdeflate       | 1.17         | matplotlib-inline                | 0.1.6    |
| intel-openmp                      | 2023.1.0  | libedit          | 3.1.20221030 | mdurl                            | 0.1.0    |
| ipykernel                         | 6.19.2    | libev            | 4.33         | minikanren                       | 1.0.3    |
| ipympl                            | 0.8.7     | libffi           | 3.4.4        | mistune                          | 2.0.4    |
| ipython                           | 7.34.0    | libgcc-ng        | 11.2.0       | mkl                              | 2023.1.0 |
| ipython_genutils                  | 0.2.0     | libgd            | 2.3.3        | mkl-service                      | 2.4.0    |
| ipywidgets                        | 7.7.2     | libgdal          | 3.4.1        | mkl_fft                          | 1.3.8    |
| isodate                           | 0.6.0     | libgfortran-ng   | 11.2.0       | mkl_random                       | 1.2.4    |
| jaraco.classes                    | 3.2.1     | libgfortran5     | 11.2.0       | mljar-supervised                 | 0.11.2   |
| jasper                            | 2.0.14    | libglu           | 9.0.0        | mmcls                            | 0.25.0   |
| jedi                              | 0.18.2    | libgomp          | 11.2.0       | mmcv-full                        | 1.6.0    |
| jeepney                           | 0.7.1     | liblvm14         | 14.0.6       | mmdet                            | 2.27.0   |
| jinja2                            | 3.1.2     | libmamba         | 1.5.1        | mmdet3d                          | 1.0.0rc5 |
| jmespath                          | 0.10.0    | libmambapy       | 1.5.1        | mmengine                         | 0.7.3    |
| joblib                            | 1.2.0     | libnetcdf        | 4.8.1        | mmsegmentation                   | 0.30.0   |
| jpeg                              | 9e        | libnghttp2       | 1.52.0       | mock                             | 4.0.3    |
| json-c                            | 0.16      | libopencl        | 4.5.2        | more-itertools                   | 8.12.0   |
| jsonpatch                         | 1.32      | libopus          | 1.3.1        | motmetrics                       | 1.1.3    |
| jsonpointer                       | 2.1       | libpng           | 1.6.39       | mpc                              | 1.1.0    |
| jsonschema                        | 4.17.3    | libpq            | 12.15        | mpfr                             | 4.0.2    |
| jupyter_client                    | 6.1.12    | libprotobuf      | 3.20.3       | mpmath                           | 1.3.0    |
| jupyter_contrib_core              | 0.4.0     | librsync         | 2.54.4       | msgpack-python                   | 1.0.3    |
| jupyter_contrib_nbextensions      | 0.5.1     | libsodium        | 1.0.18       | msrest                           | 0.6.21   |
| jupyter_core                      | 4.11.2    | libsolv          | 0.7.24       | multi-scale-deformable-attention | 1.0.0    |
| jupyter_highlight_selected_word   | 0.2.0     | libspatialindex  | 1.9.3        | multidict                        | 6.0.2    |
| jupyter_latex_envs                | 1.4.6     | libspatialite    | 4.3.0a       | multipledispatch                 | 0.6.0    |
| jupyter_nbextensions_configurator | 0.4.1     | libssh2          | 1.10.0       | munch                            | 2.5.0    |
| jupyter_server                    | 1.21.0    | libstdcxx-ng     | 11.2.0       | munkres                          | 1.1.4    |
| jupyterlab_pygments               | 0.2.2     | libtiff          | 4.5.1        | murmurhash                       | 1.0.7    |
| jupyterlab_widgets                | 3.0.3     | libtool          | 2.4.6        | mypy_extensions                  | 1.0.0    |
| jxrlib                            | 1.1       | libuuid          | 1.41.5       | nb_conda_kernels                 | 2.3.1    |
| kealib                            | 1.5.0     | libuv            | 1.44.2       | nbclassic                        | 0.5.5    |
| keepalive                         | 0.5       | libvpx           | 1.7.0        | nbclient                         | 0.7.0    |
| keras-base                        | 2.7.0     | libwebp          | 1.3.2        | nbconvert                        | 7.2.1    |
| keras-gpu                         | 2.7.0     | libwebp-base     | 1.3.2        | nbformat                         | 5.6.1    |
| keras-preprocessing               | 1.1.2     | libxcb           | 1.15         | ncurses                          | 6.4      |
| keyring                           | 23.13.1   | libxgboost       | 1.5.0        | nest-asyncio                     | 1.5.6    |
| kiwisolver                        | 1.4.4     | libxml2          | 2.10.4       | netcdf4                          | 1.6.2    |
| krb5                              | 1.20.1    | libxslt          | 1.1.37       | networkx                         | 3.1      |
|                                   |           | libzip           | 1.8.0        | neural-structured-learning       | 1.4.0    |
|                                   |           | libzopfli        | 1.0.3        |                                  |          |
|                                   |           | lightgbm         | 3.3.5        |                                  |          |

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| Library Name      | Version | Library Name          | Version      | Library Name                  | Version  |
|-------------------|---------|-----------------------|--------------|-------------------------------|----------|
| ninja             | 1.10.2  | py-opencv             | 4.5.2        | requests-oauthlib             | 1.3.0    |
| ninja-base        | 1.10.2  | py-xgboost            | 1.5.0        | requests-toolbelt             | 1.0.0    |
| nose              | 1.3.7   | pyasn1                | 0.4.8        | rich                          | 13.3.5   |
| notebook          | 6.5.4   | pyasn1-modules        | 0.2.8        | rise                          | 5.7.1    |
| notebook-shim     | 0.1.0   | pybind11              | 2.10.4       | rsa                           | 4.7.2    |
| numba             | 0.57.1  | pybind11-abi          | 4            | rtree                         | 0.9.4    |
| numexpr           | 2.8.4   | pybind11-global       | 2.10.4       | ruamel.yaml                   | 0.17.21  |
| numpy             | 1.22.3  | pycocotools           | 2.0.2        | ruamel.yaml.clib              | 0.2.6    |
| numpy-base        | 1.22.3  | pycosat               | 0.6.4        | s3transfer                    | 0.6.0    |
| nuscenes-devkit   | 1.1.3   | pyparser              | 2.21         | sacremoses                    | 0.0.43   |
| nvidia-ml-py3     | 7.352.0 | pyct                  | 0.5.0        | scikit-image                  | 0.19.3   |
| oauthlib          | 3.2.2   | pygments              | 2.14.0       | scikit-learn                  | 1.2.2    |
| onnx              | 1.13.0  | pyjwt                 | 2.4.0        | scikit-plot                   | 0.3.7    |
| onnx-tf           | 1.8.0   | pykerberos            | 1.2.1        | scipy                         | 1.9.3    |
| opencv            | 4.5.2   | pylerc                | 3.0          | seaborn                       | 0.12.2   |
| openjpeg          | 2.4.0   | pymc                  | 5.6.1        | secretstorage                 | 3.3.1    |
| openpyxl          | 3.0.10  | pyodbc                | 4.0.34       | send2trash                    | 1.8.0    |
| openssl           | 3.0.11  | pyopenssl             | 23.2.0       | sentencepiece                 | 0.1.99   |
| opt_einsum        | 3.3.0   | pyparsing             | 3.0.9        | setuptools                    | 68.0.0   |
| optuna            | 2.8.0   | pypdf                 | 3.15.0       | shap                          | 0.39.0   |
| packaging         | 23.1    | pypdf2                | 1.26.0       | shapely                       | 2.0.1    |
| pandas            | 2.0.2   | pyperclip             | 1.8.2        | simplejson                    | 3.17.6   |
| pandocfilters     | 1.5.0   | pyquaternion          | 0.9.9        | six                           | 1.16.0   |
| panel             | 0.14.3  | pyrsistent            | 0.18.0       | slicer                        | 0.0.7    |
| pango             | 1.50.7  | pyshp                 | 2.1.3        | snappy                        | 1.1.9    |
| papermill         | 2.3.4   | pysocks               | 1.7.1        | sniffio                       | 1.2.0    |
| param             | 1.13.0  | pyspnego              | 0.8.0        | snuggs                        | 1.4.7    |
| parso             | 0.8.3   | pytables              | 3.7.0        | sortedcontainers              | 2.4.0    |
| partd             | 1.4.0   | pytensor              | 2.12.3       | soupsieve                     | 2.4      |
| pathspec          | 0.9.0   | pytest                | 7.4.0        | spacy                         | 2.2.4    |
| patsy             | 0.5.3   | python                | 3.9.18       | sparglwrapper                 | 1.8.5    |
| pbr               | 5.6.0   | python-dateutil       | 2.8.2        | sqlalchemy                    | 1.4.39   |
| pcre              | 8.45    | python-fastjsonschema | 2.16.2       | sqlite                        | 3.41.2   |
| pcre2             | 10.42   | python-flatbuffers    | 1.12         | srsly                         | 1.0.5    |
| pexpect           | 4.8.0   | python-graphviz       | 0.20.1       | stack_data                    | 0.5.1    |
| pickleshare       | 0.7.5   | python-gssapi         | 1.8.1        | statsmodels                   | 0.13.5   |
| pillow            | 9.4.0   | python-sounddevice    | 0.4.4        | stevedore                     | 3.3.0    |
| pip               | 23.2.1  | python_abi            | 3.9          | sympy                         | 1.11.1   |
| pixman            | 0.40.0  | pytorch               | 1.8.2        | tabulate                      | 0.8.9    |
| plac              | 1.1.0   | pytz                  | 2023.3.post1 | tbb                           | 2021.8.0 |
| platformdirs      | 3.10.0  | pyviz_comms           | 2.3.0        | tblib                         | 1.7.0    |
| plotly            | 5.15.0  | pywavelets            | 1.4.1        | tenacity                      | 8.2.2    |
| pluggy            | 1.0.0   | pyyaml                | 6.0          | tensorboard                   | 2.7.0    |
| portaudio         | 19.6.0  | pyzmq                 | 24.0.1       | tensorboard-data-server       | 0.6.1    |
| postgresql        | 12.15   | rasterio              | 1.2.10       | tensorboard-plugin-wit        | 1.8.1    |
| preshed           | 3.0.6   | rasterstats           | 0.16.0       | tensorboardx                  | 2.2      |
| prettytable       | 2.1.0   | rdflib                | 5.0.0        | tensorflow-addons             | 0.15.0   |
| proj              | 6.2.1   | re2                   | 2022.04.01   | tensorflow-estimator          | 2.7.0    |
| prometheus_client | 0.8.0   | readline              | 8.2          | tensorflow-gpu                | 2.7.0    |
| prompt_toolkit    | 3.0.5   | regex                 | 2022.7.9     | tensorflow-hub                | 0.12.0   |
| protobuf          | 3.20.3  | reproc                | 14.2.4       | tensorflow-model-optimization | 0.7.3    |
| psutil            | 5.9.0   | reproc-cpp            | 14.2.4       | termcolor                     | 2.1.0    |
| ptyprocess        | 0.7.0   | requests              | 2.31.0       | terminado                     | 0.17.1   |
| pure_eval         | 0.2.2   | requests-gssapi       | 1.2.3        | terminaltables                | 3.1.0    |
| py-boost          | 1.79.0  | requests-kerberos     | 0.12.0       |                               |          |

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| Library Name       | Version   |
|--------------------|-----------|
| textwrap3          | 0.9.2     |
| tflite-model-maker | 0.3.4     |
| tflite-support     | 0.4.1     |
| thinc              | 7.4.0     |
| threadpoolctl      | 2.2.0     |
| tifffile           | 2023.4.12 |
| tiledb             | 2.3.3     |
| timm               | 0.4.12    |
| tinycss2           | 1.1.1     |
| tk                 | 8.6.12    |
| tokenizers         | 0.10.1    |
| tomli              | 2.0.1     |
| toolz              | 0.12.0    |
| torch-cluster      | 1.5.9     |
| torch-geometric    | 1.7.2     |
| torch-scatter      | 2.0.7     |
| torch-sparse       | 0.6.10    |
| torch-spline-conv  | 1.2.1     |
| torchvision        | 0.9.2     |
| tornado            | 6.3.2     |
| tqdm               | 4.65.0    |
| traitlets          | 5.5.0     |
| transformers       | 4.14.1    |
| trimesh            | 2.35.39   |
| typingguard        | 2.13.0    |
| typing-extensions  | 4.7.1     |
| typing_extensions  | 4.7.1     |
| tzdata             | 2023c     |
| ujson              | 5.4.0     |
| unixodbc           | 2.3.11    |
| urllib3            | 1.26.16   |
| wasabi             | 0.9.1     |
| wcwidth            | 0.2.5     |
| webencodings       | 0.5.1     |
| websocket-client   | 0.58.0    |
| werkzeug           | 2.2.3     |
| wheel              | 0.41.2    |
| widgetsnbextension | 3.6.1     |
| wordcloud          | 1.8.1     |
| wrapt              | 1.14.1    |
| xarray             | 2023.6.0  |
| xarray-einstats    | 0.6.0     |
| xerces-c           | 3.2.4     |
| xgboost            | 1.5.0     |
| xlrd               | 2.0.1     |
| xlwt               | 1.3.0     |
| xmltodict          | 0.12.0    |
| xz                 | 5.4.2     |
| yaml               | 0.2.5     |
| yaml-cpp           | 0.7.0     |
| yapf               | 0.31.0    |
| yaml               | 1.8.1     |
| zeromq             | 4.3.4     |
| zfp                | 1.0.0     |
| zict               | 3.0.0     |

| Library Name | Version |
|--------------|---------|
| zipp         | 3.11.0  |
| zlib         | 1.2.13  |
| zstandard    | 0.19.0  |
| zstd         | 1.5.5   |

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# ArcGIS Notebooks Runtime

## Python 3 Standard (Windows Containers) | Version 9.0

### RELEASED

October 2023

### TOTAL PACKAGES

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### SUPPORTED IN

ArcGIS Enterprise 11.2

In ArcGIS Enterprise you can customize your notebook environment, learn more at [Extend Notebook Runtime](#).

| Library Name         | Version   | Library Name       | Version   | Library Name              | Version   |
|----------------------|-----------|--------------------|-----------|---------------------------|-----------|
| _py-xgboost-mutex    | 2.0       | charls             | 2.2.0     | font-ttf-dejavu-sans-mono | 2.37      |
| _tflow_select        | 2.7.0     | charset-normalizer | 2.0.4     | font-ttf-inconsolata      | 2.001     |
| absl-py              | 1.4.0     | click              | 8.0.4     | font-ttf-source-code-pro  | 2.030     |
| addict               | 2.4.0     | click-plugins      | 1.1.1     | font-ttf-ubuntu           | 0.83      |
| affine               | 2.3.0     | cliff              | 3.8.0     | fontconfig                | 2.14.1    |
| aiohttp              | 3.8.5     | cligj              | 0.7.2     | fonts-anaconda            | 1         |
| aiosignal            | 1.2.0     | cloudpickle        | 2.2.1     | fonts-conda-ecosystem     | 1         |
| albumations          | 1.0.3     | cmaes              | 0.8.2     | fonttools                 | 4.25.0    |
| alembic              | 1.8.1     | cmd2               | 2.4.2     | freetype                  | 2.12.1    |
| altair               | 4.1.0     | colorama           | 0.4.6     | fribidi                   | 1.0.10    |
| ansiwrap             | 0.8.4     | colorcet           | 3.0.1     | frozenlist                | 1.3.3     |
| anyio                | 3.5.0     | colorlog           | 5.0.1     | fsspec                    | 2023.9.2  |
| aom                  | 3.6.0     | colour             | 0.1.5     | future                    | 0.18.2    |
| appdirs              | 1.4.4     | comm               | 0.1.4     | gast                      | 0.4.0     |
| arcgis               | 2.2.0.1   | contourpy          | 1.0.5     | gdal                      | 3.7.0     |
| arcgispro            | 3.2       | cppzmq             | 4.9.0     | geomet                    | 1.0.0     |
| argon2-cffi          | 21.3.0    | cryptography       | 41.0.3    | geos                      | 3.5.0     |
| argon2-cffi-bindings | 21.2.0    | cudatoolkit        | 11.1.1    | getopt-win32              | 0.1       |
| arrow-cpp            | 12.0.0    | cudnn              | 8.1.0.77  | giflib                    | 5.2.1     |
| arviz                | 0.11.2    | cycler             | 0.11.0    | glib                      | 2.69.1    |
| astor                | 0.8.1     | cymem              | 2.0.6     | google-auth               | 2.22.0    |
| asttokens            | 2.0.5     | cython             | 0.29.33   | google-auth-oauthlib      | 0.4.1     |
| astunparse           | 1.6.3     | cython-blis        | 0.4.1     | google-pasta              | 0.2.0     |
| async-timeout        | 4.0.2     | cytoolz            | 0.12.0    | googledrivedownloader     | 0.4       |
| attrs                | 23.1.0    | dask               | 2023.3.2  | graphite2                 | 1.3.14    |
| azure-core           | 1.12.0    | dask-core          | 2023.3.2  | graphviz                  | 2.38      |
| azure-storage-blob   | 12.8.0    | dask-glm           | 0.2.0     | greenlet                  | 2.0.1     |
| babel                | 2.11.0    | dask-ml            | 2022.5.27 | grpcio                    | 1.42.0    |
| backcall             | 0.2.0     | dataclasses        | 0.8       | gts                       | 0.7.6     |
| beautifulsoup4       | 4.12.2    | dav1d              | 1.2.1     | h3-py                     | 3.7.3     |
| black                | 22.1.0    | debugpy            | 1.6.7     | h5py                      | 3.9.0     |
| blas                 | 1.0       | decorator          | 5.1.1     | hdf5                      | 1.10.6    |
| bleach               | 4.1.0     | defusedxml         | 0.7.1     | heapdict                  | 1.0.1     |
| blinker              | 1.6.2     | descartes          | 1.1.0     | holoviews                 | 1.15.4    |
| blosc                | 1.21.3    | dill               | 0.3.6     | html5lib                  | 1.1       |
| bokeh                | 2.4.3     | distributed        | 2023.3.2  | icc_rt                    | 2022.2.1  |
| boost                | 1.73.0    | dm-tree            | 0.1.7     | icu                       | 68.1      |
| boto3                | 1.24.28   | dpcpp-cpp-rt       | 2022.2.1  | idna                      | 3.4       |
| botocore             | 1.27.59   | dtreeviz           | 1.3.7     | imagecodecs               | 2023.1.23 |
| bottleneck           | 1.3.5     | einops             | 0.3.2     | imageio                   | 2.31.4    |
| brtoli               | 1.0.9     | ensemble-boxes     | 1.0.8     | imgaug                    | 0.4.0     |
| brtoli-bin           | 1.0.9     | entrypoints        | 0.4       | importlib-metadata        | 6.0.0     |
| brtolipty            | 0.7.0     | et_xmlfile         | 1.1.0     | importlib_metadata        | 6.0.0     |
| bzip2                | 1.0.8     | exceptiongroup     | 1.0.0rc9  | iniconfig                 | 1.1.1     |
| c-ares               | 1.19.1    | executing          | 0.8.3     | inplace-abn               | 1.1.0     |
| cachetools           | 4.2.2     | expat              | 2.5.0     | intel-cmplr-lib-rt        | 2022.2.1  |
| cairo                | 1.16.0    | fairlearn          | 0.8.0     | intel-fortran-rt          | 2022.2.1  |
| catalogue            | 1.0.0     | fastai             | 1.0.63    | intel-openmp              | 2022.2.1  |
| catboost             | 0.26      | fastprogress       | 0.2.3     | ipykernel                 | 6.19.2    |
| category_encoders    | 2.2.2     | fasttext           | 0.9.2     | ipynb                     | 0.8.7     |
| certifi              | 2023.7.22 | filelock           | 3.9.0     | ipython                   | 7.34.0    |
| cffi                 | 1.15.1    | fiona              | 1.8.22    | ipython_genutils          | 0.2.0     |
| cfitsio              | 3.470     | fire               | 0.4.0     | ipywidgets                | 7.7.2     |
| cftime               | 1.6.2     | flake8             | 4.0.1     |                           |           |
| chardet              | 4.0.0     | flatbuffers        | 2.0.0     |                           |           |

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## Python 3 Standard (Windows Containers) | Version 9.0

| Library Name                      | Version    | Library Name                     | Version  | Library Name               | Version   |
|-----------------------------------|------------|----------------------------------|----------|----------------------------|-----------|
| isodate                           | 0.6.0      | libthrift                        | 0.18.1   | nbformat                   | 5.6.1     |
| jaraco.classes                    | 3.2.1      | libtiff                          | 4.5.1    | nest-asyncio               | 1.5.6     |
| jdcal                             | 1.4.1      | libutf8proc                      | 2.8.0    | netcdf4                    | 1.6.4     |
| jedi                              | 0.18.2     | libuv                            | 1.40.0   | networkx                   | 3.1       |
| jinja2                            | 3.0.3      | libwebp                          | 1.3.2    | neural-structured-learning | 1.4.0     |
| jmespath                          | 0.10.0     | libwebp-base                     | 1.3.2    | ninja                      | 1.10.2    |
| joblib                            | 1.2.0      | libxgboost                       | 1.5.0    | ninja-base                 | 1.10.2    |
| jpeg                              | 9e         | libxml2                          | 2.10.4   | nlohmann_json              | 3.11.2    |
| json5                             | 0.9.5      | libxslt                          | 1.1.37   | nose                       | 1.3.7     |
| jsonschema                        | 4.17.3     | libzopfli                        | 1.0.3    | notebook                   | 6.4.12    |
| jupyter_client                    | 6.1.12     | lightgbm                         | 3.3.5    | notebook-shim              | 0.1.0     |
| jupyter_console                   | 6.4.2      | llvmlite                         | 0.40.0   | ntlm-auth                  | 1.4.0     |
| jupyter_contrib_core              | 0.4.0      | locket                           | 1.0.0    | numba                      | 0.57.1    |
| jupyter_contrib_nbextensions      | 0.6.0      | lxml                             | 4.9.1    | numexpr                    | 2.8.4     |
| jupyter_core                      | 4.11.2     | lz4                              | 4.3.2    | numpy                      | 1.22.4    |
| jupyter_highlight_selected_word   | 0.2.0      | lz4-c                            | 1.9.4    | numpy-base                 | 1.22.4    |
| jupyter_latex_envs                | 1.4.6      | lzo                              | 2.10     | nuscenes-devkit            | 1.1.3     |
| jupyter_nbextensions_configurator | 0.4.1      | mako                             | 1.2.3    | nvml-py3                   | 7.352.0   |
| jupyter_server                    | 1.21.0     | markdown                         | 3.4.1    | oauthlib                   | 3.2.0     |
| jupyterlab                        | 3.4.8      | markdown-it-py                   | 2.2.0    | olefile                    | 0.46      |
| jupyterlab_pygments               | 0.2.2      | markupsafe                       | 2.1.1    | onnx                       | 1.9.0     |
| jupyterlab_server                 | 2.16.0     | matplotlib                       | 3.6.3    | onnx-tf                    | 1.8.0     |
| jupyterlab_widgets                | 3.0.3      | matplotlib-base                  | 3.6.3    | opencv                     | 4.5.2     |
| keepalive                         | 0.5        | matplotlib-inline                | 0.1.6    | openjpeg                   | 2.4.0     |
| keras                             | 2.7.0      | mccabe                           | 0.7.0    | openpyxl                   | 3.0.10    |
| keras-base                        | 2.7.0      | mdurl                            | 0.1.0    | openssl                    | 3.0.10    |
| keras-gpu                         | 2.7.0      | mistune                          | 2.0.4    | opt_einsum                 | 3.3.0     |
| keras-preprocessing               | 1.1.2      | mkl                              | 2022.2.1 | optuna                     | 2.8.0     |
| keyring                           | 23.13.1    | mkl-service                      | 2.4.0    | orc                        | 1.8.3     |
| kiwisolver                        | 1.4.4      | mkl_fft                          | 1.3.1    | packaging                  | 21.3      |
| krb5                              | 1.21.1     | mkl_random                       | 1.2.2    | pandas                     | 2.0.2     |
| laspy                             | 1.7.0      | mljar-supervised                 | 0.11.2   | pandocfilters              | 1.5.0     |
| lcms2                             | 2.12       | mmcls                            | 0.25.0   | panel                      | 0.14.3    |
| lerc                              | 3.0        | mmcv-full                        | 1.6.0    | papermill                  | 2.3.4     |
| libabseil                         | 20230125.3 | mmdet                            | 2.27.0   | param                      | 1.13.0    |
| libaec                            | 1.0.4      | mmdet3d                          | 1.0.0rc5 | parso                      | 0.8.3     |
| libavif                           | 0.11.1     | mmengine                         | 0.7.3    | partd                      | 1.4.0     |
| libboost                          | 1.73.0     | mmsegmentation                   | 0.30.0   | pathspec                   | 0.9.0     |
| libbrotlicommon                   | 1.0.9      | mock                             | 4.0.3    | patsy                      | 0.5.3     |
| libbrotlidec                      | 1.0.9      | more-itertools                   | 8.12.0   | pbr                        | 5.6.0     |
| libbrotlienc                      | 1.0.9      | motmetrics                       | 1.1.3    | pcr                        | 8.45      |
| libcurl                           | 7.86.0     | mpmath                           | 1.3.0    | pefile                     | 2022.5.30 |
| libdeflate                        | 1.17       | msgpack-python                   | 1.0.3    | pexpect                    | 4.8.0     |
| libffi                            | 3.4.4      | msrest                           | 0.6.21   | pickleshare                | 0.7.5     |
| libgd                             | 2.3.3      | multi-scale-deformable-attention | 1.0.0    | pillow                     | 9.5.0     |
| libiconv                          | 1.16       | multidict                        | 6.0.2    | pip                        | 23.2.1    |
| libnghttp2                        | 1.50.0     | multiplatformdispatch            | 0.6.0    | pixman                     | 0.40.0    |
| libopencv                         | 4.5.2      | munch                            | 2.5.0    | plac                       | 1.1.0     |
| libpng                            | 1.6.39     | munkres                          | 1.1.4    | platformdirs               | 3.10.0    |
| libprotobuf                       | 3.21.12    | murmurhash                       | 1.0.7    | plotly                     | 5.5.0     |
| libsodium                         | 1.0.18     | mypy_extensions                  | 0.4.3    | pluggy                     | 1.0.0     |
| libspatialindex                   | 1.9.3      | nbclassic                        | 0.4.8    | portaudio                  | 19.6.0    |
|                                   |            | nbclient                         | 0.7.0    | reshed                     | 3.0.6     |
|                                   |            | nbconvert                        | 7.2.1    | prettytable                | 2.1.0     |

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| Library Name             | Version | Library Name            | Version    | Library Name                  | Version     |
|--------------------------|---------|-------------------------|------------|-------------------------------|-------------|
| pro_notebook_integration | 3.2     | pywin32-security        | 306        | tblib                         | 1.7.0       |
| proj                     | 8.2.1   | pywinpty                | 2.0.1      | tenacity                      | 8.2.2       |
| prometheus_client        | 0.8.0   | pyyaml                  | 6.0        | tensorboard                   | 2.7.0       |
| prompt_toolkit           | 3.0.5   | pyzmq                   | 25.0.2     | tensorboard-data-server       | 0.6.1       |
| protobuf                 | 3.21.12 | rasterio                | 1.2.10     | tensorboard-plugin-wit        | 1.8.1       |
| psutil                   | 5.9.0   | rasterstats             | 0.16.0     | tensorboardx                  | 2.2         |
| ptyprocess               | 0.7.0   | rdflib                  | 5.0.0      | tensorflow-addons             | 0.15.0      |
| pure_eval                | 0.2.2   | re2                     | 2023.06.02 | tensorflow-base               | 2.7.0       |
| py-boost                 | 1.73.0  | regex                   | 2023.10.3  | tensorflow-estimator          | 2.7.0       |
| py-opencv                | 4.5.2   | requests                | 2.31.0     | tensorflow-gpu                | 2.7.0       |
| py-xgboost               | 1.5.0   | requests-gssapi         | 1.2.3      | tensorflow-hub                | 0.12.0      |
| pyarrow                  | 12.0.0  | requests-kerberos       | 0.12.0     | tensorflow-model-optimization | 0.7.3       |
| pyasn1                   | 0.4.8   | requests-negotiate-sspi | 0.5.3      | termcolor                     | 2.1.0       |
| pyasn1-modules           | 0.2.8   | requests-oauthlib       | 1.3.0      | terminado                     | 0.17.1      |
| pybind11                 | 2.10.4  | requests-toolbelt       | 1.0.0      | terminaltables                | 3.1.0       |
| pybind11_json            | 0.2.13  | rich                    | 13.3.5     | testpath                      | 0.5.0       |
| pycocotools              | 2.0.2   | rise                    | 5.7.1      | textwrap3                     | 0.9.2       |
| pycodestyle              | 2.8.0   | rsa                     | 4.7.2      | tfliite-model-maker           | 0.3.4       |
| pycparser                | 2.21    | rtree                   | 0.9.4      | tfliite-support               | 0.4.1       |
| pyct                     | 0.5.0   | ruamel.yaml             | 0.17.21    | theano-pymc                   | 1.1.2       |
| pyee                     | 8.2.2   | ruamel.yaml.clib        | 0.2.6      | thinc                         | 7.4.0       |
| pyflakes                 | 2.4.0   | s3transfer              | 0.6.0      | threadpoolctl                 | 2.2.0       |
| pygments                 | 2.14.0  | sacremoses              | 0.0.43     | tifffile                      | 2023.4.12   |
| pyjwt                    | 2.4.0   | saspy                   | 4.3.2      | timm                          | 0.4.12      |
| pylerc                   | 4.0     | scikit-image            | 0.19.3     | tinycss2                      | 1.1.1       |
| pymc3                    | 3.11.4  | scikit-learn            | 1.0.2      | tokenizers                    | 0.10.1      |
| pyodbc                   | 4.0.34  | scikit-plot             | 0.3.7      | toml                          | 0.10.2      |
| pyopenssl                | 23.2.0  | scipy                   | 1.7.3      | tomli                         | 2.0.1       |
| pyarsing                 | 3.0.9   | seaborn                 | 0.12.2     | toolz                         | 0.12.0      |
| pypdf                    | 3.15.0  | semver                  | 2.13.0     | torch-cluster                 | 1.5.9       |
| pypdf2                   | 1.26.0  | send2trash              | 1.8.0      | torch-geometric               | 1.7.2       |
| pyperclip                | 1.8.2   | sentencepiece           | 0.1.99     | torch-scatter                 | 2.0.7       |
| pyppeteer                | 1.0.2   | setuptools              | 67.7.2     | torch-sparse                  | 0.6.10      |
| pyquaternion             | 0.9.9   | shap                    | 0.39.0     | torch-spline-conv             | 1.2.1       |
| pyrsistent               | 0.18.0  | shapely                 | 1.7.1      | torchvision                   | 0.9.2       |
| pyshp                    | 2.1.3   | simplejson              | 3.17.6     | tornado                       | 6.3.2       |
| pysocks                  | 1.7.1   | six                     | 1.16.0     | tqdm                          | 4.65.0      |
| pyspnego                 | 0.9.2   | slicer                  | 0.0.7      | traitlets                     | 5.5.0       |
| pytables                 | 3.7.0   | snappy                  | 1.1.10     | transformers                  | 4.5.1       |
| pytest                   | 7.2.0   | sniffio                 | 1.2.0      | trimesh                       | 2.35.39     |
| python                   | 3.9.18  | snuggs                  | 1.4.7      | typed-ast                     | 1.4.3       |
| python-certifi-win32     | 1.6     | sortedcontainers        | 2.4.0      | typeguard                     | 2.13.0      |
| python-dateutil          | 2.8.2   | soupsieve               | 2.5        | typing                        | 3.10.0.0    |
| python-fastjsonschema    | 2.16.2  | spacy                   | 2.2.4      | typing-extensions             | 4.7.1       |
| python-flatbuffers       | 1.12    | sparqlwrapper           | 1.8.5      | typing_extensions             | 4.7.1       |
| python-graphviz          | 0.20.1  | sqlalchemy              | 1.4.39     | tzdata                        | 2023c       |
| python-gssapi            | 1.8.2   | sqlite                  | 3.41.2     | ujson                         | 5.4.0       |
| python-sounddevice       | 0.4.4   | srsly                   | 1.0.5      | urllib3                       | 1.26.16     |
| python_abi               | 3.9     | stack_data              | 0.5.1      | vc                            | 14.2        |
| pytorch                  | 1.8.2   | statsmodels             | 0.14.0     | vs2015_runtime                | 14.27.29016 |
| pytz                     | 2022.6  | stevedore               | 3.3.0      | wasabi                        | 0.9.1       |
| pyviz_comms              | 2.3.0   | swat                    | 1.13.2     | wcwidth                       | 0.2.5       |
| pywavelets               | 1.4.1   | sympy                   | 1.9        | webencodings                  | 0.5.1       |
| pywin32-ctypes           | 0.2.0   | tabulate                | 0.8.9      |                               |             |
|                          |         | tbb                     | 2021.8.0   |                               |             |

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| Library Name       | Version  |
|--------------------|----------|
| websocket-client   | 0.58.0   |
| websockets         | 10.4     |
| werkzeug           | 2.2.3    |
| wheel              | 0.37.1   |
| widgetsnbextension | 3.6.1    |
| win_inet_pton      | 1.1.0    |
| wincertstore       | 0.2      |
| winkerberos        | 0.8.0    |
| winpty             | 0.4.3    |
| wordcloud          | 1.8.1    |
| wrapt              | 1.14.1   |
| x86cpu             | 0.4      |
| xarray             | 2023.6.0 |
| xeus               | 3.0.5    |
| xeus-python        | 0.15.8   |
| xeus-zmq           | 1.0.2    |
| xgboost            | 1.5.0    |
| xlrd               | 2.0.1    |
| xlwt               | 1.3.0    |
| xmltodict          | 0.12.0   |
| xtl                | 0.7.5    |
| xz                 | 5.4.2    |
| yaml               | 0.2.5    |
| yapf               | 0.31.0   |
| yaml               | 1.8.1    |
| zeromq             | 4.3.4    |
| zfp                | 1.0.0    |
| zict               | 3.0.0    |
| zipp               | 3.11.0   |
| zlib               | 1.2.13   |
| zlib-ng            | 2.0.7    |
| zstd               | 1.5.5    |

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## Python 3 Advanced (Windows Containers) | Version 9.0

### RELEASED

October 2023

### TOTAL PACKAGES

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### SUPPORTED IN

ArcGIS Enterprise 11.2

The use of the Advanced and/or Advanced with GPU runtime consumes credits within ArcGIS Online. See ArcGIS Online [credits by capability](#) for more details.

In ArcGIS Enterprise, you can customize your notebook environment, learn more at [Extend Notebook Runtime](#).

| Library Name         | Version   | Library Name       | Version   | Library Name              | Version   |
|----------------------|-----------|--------------------|-----------|---------------------------|-----------|
| _py-xgboost-mutex    | 2.0       | cftime             | 1.6.2     | flake8                    | 4.0.1     |
| _tflow_select        | 2.7.0     | chardet            | 4.0.0     | flatbuffers               | 2.0.0     |
| absl-py              | 1.4.0     | charls             | 2.2.0     | font-ttf-dejavu-sans-mono | 2.37      |
| addict               | 2.4.0     | charset-normalizer | 2.0.4     | font-ttf-inconsolata      | 2.001     |
| affine               | 2.3.0     | click              | 8.0.4     | font-ttf-source-code-pro  | 2.030     |
| aiohttp              | 3.8.5     | click-plugins      | 1.1.1     | font-ttf-ubuntu           | 0.83      |
| aiosignal            | 1.2.0     | cliff              | 3.8.0     | fontconfig                | 2.14.1    |
| albumations          | 1.0.3     | cligj              | 0.7.2     | fonts-anaconda            | 1         |
| alembic              | 1.8.1     | cloudpickle        | 2.2.1     | fonts-conda-ecosystem     | 1         |
| altair               | 4.1.0     | cmaes              | 0.8.2     | fonttools                 | 4.25.0    |
| ansiwrap             | 0.8.4     | cmd2               | 2.4.2     | freetype                  | 2.12.1    |
| anyio                | 3.5.0     | colorama           | 0.4.6     | fribidi                   | 1.0.10    |
| aom                  | 3.6.0     | colorcet           | 3.0.1     | frozenlist                | 1.3.3     |
| appdirs              | 1.4.4     | colorlog           | 5.0.1     | fspec                     | 2023.9.2  |
| arcgis               | 2.2.0.1   | colour             | 0.1.5     | future                    | 0.18.2    |
| arcgispro            | 3.2       | comm               | 0.1.4     | gast                      | 0.4.0     |
| arcpy                | 3.2       | contourpy          | 1.0.5     | gdal                      | 3.7.0     |
| arcpy-base           | 3.2       | cppzmq             | 4.9.0     | geomnet                   | 1.0.0     |
| argon2-cffi          | 21.3.0    | cryptography       | 41.0.3    | geos                      | 3.5.0     |
| argon2-cffi-bindings | 21.2.0    | cuda-toolkit       | 11.1.1    | getopt-win32              | 0.1       |
| arrow-cpp            | 12.0.0    | cuda-nn            | 8.1.0.77  | giflib                    | 5.2.1     |
| arviz                | 0.11.2    | cycler             | 0.11.0    | glib                      | 2.69.1    |
| astor                | 0.8.1     | cymem              | 2.0.6     | google-auth               | 2.22.0    |
| asttokens            | 2.0.5     | cython             | 0.29.33   | google-auth-oauthlib      | 0.4.1     |
| astunparse           | 1.6.3     | cython-blis        | 0.4.1     | google-pasta              | 0.2.0     |
| async-timeout        | 4.0.2     | cytoolz            | 0.12.0    | googledrivedownloader     | 0.4       |
| attrs                | 23.1.0    | dask               | 2023.3.2  | graphite2                 | 1.3.14    |
| azure-core           | 1.12.0    | dask-core          | 2023.3.2  | graphviz                  | 2.38      |
| azure-storage-blob   | 12.8.0    | dask-glm           | 0.2.0     | greenlet                  | 2.0.1     |
| babel                | 2.11.0    | dask-ml            | 2022.5.27 | grpcio                    | 1.42.0    |
| backcall             | 0.2.0     | dataclasses        | 0.8       | gts                       | 0.7.6     |
| beautifulsoup4       | 4.12.2    | dav1d              | 1.2.1     | h3-py                     | 3.7.3     |
| black                | 22.1.0    | debugpy            | 1.6.7     | h5py                      | 3.9.0     |
| blas                 | 1.0       | decorator          | 5.1.1     | hdf5                      | 1.10.6    |
| bleach               | 4.1.0     | defusedxml         | 0.7.1     | heapdict                  | 1.0.1     |
| blinker              | 1.6.2     | descartes          | 1.1.0     | holoviews                 | 1.15.4    |
| blosc                | 1.21.3    | dill               | 0.3.6     | html5lib                  | 1.1       |
| bokeh                | 2.4.3     | distributed        | 2023.3.2  | icc-rt                    | 2022.2.1  |
| boost                | 1.73.0    | dm-tree            | 0.1.7     | icu                       | 68.1      |
| boto3                | 1.24.28   | dpcpp-cpp-rt       | 2022.2.1  | idna                      | 3.4       |
| botocore             | 1.27.59   | dtreeviz           | 1.3.7     | imagecodecs               | 2023.1.23 |
| bottleneck           | 1.3.5     | einops             | 0.3.2     | imageio                   | 2.31.4    |
| brtli                | 1.0.9     | ensemble-boxes     | 1.0.8     | imgaug                    | 0.4.0     |
| brtli-bin            | 1.0.9     | entrypoints        | 0.4       | importlib-metadata        | 6.0.0     |
| brtli-py             | 0.7.0     | et-xmlfile         | 1.1.0     | importlib-metadata        | 6.0.0     |
| bzip2                | 1.0.8     | exceptiongroup     | 1.0.0rc9  | iniconfig                 | 1.1.1     |
| c-ares               | 1.19.1    | executing          | 0.8.3     | inplace-abn               | 1.1.0     |
| cachetools           | 4.2.2     | expat              | 2.5.0     | intel-cmplr-lib-rt        | 2022.2.1  |
| cairo                | 1.16.0    | fairlearn          | 0.8.0     | intel-fortran-rt          | 2022.2.1  |
| catalogue            | 1.0.0     | fastai             | 1.0.63    | intel-openmp              | 2022.2.1  |
| catboost             | 0.26      | fastprogress       | 0.2.3     | ipykernel                 | 6.19.2    |
| category-encoders    | 2.2.2     | fasttext           | 0.9.2     | ipympl                    | 0.8.7     |
| certifi              | 2023.7.22 | filelock           | 3.9.0     | ipython                   | 7.34.0    |
| cffi                 | 1.15.1    | fiona              | 1.8.22    | ipython_genutils          | 0.2.0     |
| cfitsio              | 3.470     | fire               | 0.4.0     |                           |           |



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| Library Name                      | Version    | Library Name                     | Version  | Library Name               | Version   |
|-----------------------------------|------------|----------------------------------|----------|----------------------------|-----------|
| ipywidgets                        | 7.7.2      | libspatialindex                  | 1.9.3    | nbconvert                  | 7.2.1     |
| isodate                           | 0.6.0      | libthrift                        | 0.18.1   | nbformat                   | 5.6.1     |
| jaraco.classes                    | 3.2.1      | libtiff                          | 4.5.1    | nest-asyncio               | 1.5.6     |
| jdcal                             | 1.4.1      | libutf8proc                      | 2.8.0    | netcdf4                    | 1.6.4     |
| jedi                              | 0.18.2     | libuv                            | 1.40.0   | networkx                   | 3.1       |
| jinja2                            | 3.0.3      | libwebp                          | 1.3.2    | neural-structured-learning | 1.4.0     |
| jmespath                          | 0.10.0     | libwebp-base                     | 1.3.2    | ninja                      | 1.10.2    |
| joblib                            | 1.2.0      | libxgboost                       | 1.5.0    | ninja-base                 | 1.10.2    |
| jpeg                              | 9e         | libxml2                          | 2.10.4   | nlohmann_json              | 3.11.2    |
| json5                             | 0.9.5      | libxslt                          | 1.1.37   | nose                       | 1.3.7     |
| jsonschema                        | 4.17.3     | libzopfli                        | 1.0.3    | notebook                   | 6.4.12    |
| jupyter_client                    | 6.1.12     | lightgbm                         | 3.3.5    | notebook-shim              | 0.1.0     |
| jupyter_console                   | 6.4.2      | llvmlite                         | 0.40.0   | ntlm-auth                  | 1.4.0     |
| jupyter_contrib_core              | 0.4.0      | locket                           | 1.0.0    | numba                      | 0.57.1    |
| jupyter_contrib_nbextensions      | 0.6.0      | lxml                             | 4.9.1    | numexpr                    | 2.8.4     |
| jupyter_core                      | 4.11.2     | lz4                              | 4.3.2    | numpy                      | 1.22.4    |
| jupyter_highlight_selected_word   | 0.2.0      | lz4-c                            | 1.9.4    | numpy-base                 | 1.22.4    |
| jupyter_latex_envs                | 1.4.6      | lzo                              | 2.10     | nuscenes-devkit            | 1.1.3     |
| jupyter_nbextensions_configurator | 0.4.1      | mako                             | 1.2.3    | nvidia-ml-py3              | 7.352.0   |
| jupyter_server                    | 1.21.0     | markdown                         | 3.4.1    | oauthlib                   | 3.2.0     |
| jupyterlab                        | 3.4.8      | markdown-it-py                   | 2.2.0    | olefile                    | 0.46      |
| jupyterlab_pygments               | 0.2.2      | markupsafe                       | 2.1.1    | onnx                       | 1.9.0     |
| jupyterlab_server                 | 2.16.0     | matplotlib                       | 3.6.3    | onnx-tf                    | 1.8.0     |
| jupyterlab_widgets                | 3.0.3      | matplotlib-base                  | 3.6.3    | opencv                     | 4.5.2     |
| keepalive                         | 0.5        | matplotlib-inline                | 0.1.6    | openjpeg                   | 2.4.0     |
| keras                             | 2.7.0      | mccabe                           | 0.7.0    | openpyxl                   | 3.0.10    |
| keras-base                        | 2.7.0      | mdurl                            | 0.1.0    | openssl                    | 3.0.10    |
| keras-gpu                         | 2.7.0      | mistune                          | 2.0.4    | opt_einsum                 | 3.3.0     |
| keras-preprocessing               | 1.1.2      | mkl                              | 2022.2.1 | optuna                     | 2.8.0     |
| keyring                           | 23.13.1    | mkl-service                      | 2.4.0    | orc                        | 1.8.3     |
| kiwisolver                        | 1.4.4      | mkl_fft                          | 1.3.1    | packaging                  | 21.3      |
| krb5                              | 1.21.1     | mkl_random                       | 1.2.2    | pandas                     | 2.0.2     |
| laspy                             | 1.7.0      | mljar-supervised                 | 0.11.2   | pandocfilters              | 1.5.0     |
| lcms2                             | 2.12       | mmcls                            | 0.25.0   | panel                      | 0.14.3    |
| lerc                              | 3.0        | mmcv-full                        | 1.6.0    | papermill                  | 2.3.4     |
| libabseil                         | 20230125.3 | mmdet                            | 2.27.0   | param                      | 1.13.0    |
| libaec                            | 1.0.4      | mmdet3d                          | 1.0.0rc5 | parso                      | 0.8.3     |
| libavif                           | 0.11.1     | mmengine                         | 0.7.3    | partd                      | 1.4.0     |
| libboost                          | 1.73.0     | mmsegmentation                   | 0.30.0   | pathspec                   | 0.9.0     |
| libbrotlicommon                   | 1.0.9      | mock                             | 4.0.3    | patsy                      | 0.5.3     |
| libbrotlidec                      | 1.0.9      | more-itertools                   | 8.12.0   | pbr                        | 5.6.0     |
| libbrotlienc                      | 1.0.9      | motmetrics                       | 1.1.3    | pcre                       | 8.45      |
| libcurl                           | 7.86.0     | mpmath                           | 1.3.0    | pefile                     | 2022.5.30 |
| libdeflate                        | 1.17       | msgpack-python                   | 1.0.3    | pexpect                    | 4.8.0     |
| libffi                            | 3.4.4      | msrest                           | 0.6.21   | pickleshare                | 0.7.5     |
| libgd                             | 2.3.3      | multi-scale-deformable-attention | 1.0.0    | pillow                     | 9.5.0     |
| libiconv                          | 1.16       | multidict                        | 6.0.2    | pip                        | 23.2.1    |
| libnghttp2                        | 1.50.0     | multipliedispatch                | 0.6.0    | pixman                     | 0.40.0    |
| libopencv                         | 4.5.2      | munch                            | 2.5.0    | plac                       | 1.1.0     |
| libpng                            | 1.6.39     | munkres                          | 1.1.4    | platformdirs               | 3.10.0    |
| libprotobuf                       | 3.21.12    | murmurhash                       | 1.0.7    | plotly                     | 5.5.0     |
| libsodium                         | 1.0.18     | mypy_extensions                  | 0.4.3    | pluggy                     | 1.0.0     |
|                                   |            | nbclassic                        | 0.4.8    | portaudio                  | 19.6.0    |
|                                   |            | nbclient                         | 0.7.0    | preshed                    | 3.0.6     |

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|--------------------------|---------|-------------------------|------------|-------------------------------|-------------|
| prettytable              | 2.1.0   | pywin32-ctypes          | 0.2.0      | tbb                           | 2021.8.0    |
| pro_notebook_integration | 3.2     | pywin32-security        | 306        | tblib                         | 1.7.0       |
| proj                     | 8.2.1   | pywinpty                | 2.0.1      | tenacity                      | 8.2.2       |
| prometheus_client        | 0.8.0   | pyyaml                  | 6.0        | tensorboard                   | 2.7.0       |
| prompt_toolkit           | 3.0.5   | pyzmq                   | 25.0.2     | tensorboard-data-server       | 0.6.1       |
| protobuf                 | 3.21.12 | rasterio                | 1.2.10     | tensorboard-plugin-wit        | 1.8.1       |
| psutil                   | 5.9.0   | rasterstats             | 0.16.0     | tensorboardx                  | 2.2         |
| ptyprocess               | 0.7.0   | rdflib                  | 5.0.0      | tensorflow-addons             | 0.15.0      |
| pure_eval                | 0.2.2   | re2                     | 2023.06.02 | tensorflow-base               | 2.7.0       |
| py-boost                 | 1.73.0  | regex                   | 2023.10.3  | tensorflow-estimator          | 2.7.0       |
| py-opencv                | 4.5.2   | requests                | 2.31.0     | tensorflow-gpu                | 2.7.0       |
| py-xgboost               | 1.5.0   | requests-gssapi         | 1.2.3      | tensorflow-hub                | 0.12.0      |
| pyarrow                  | 12.0.0  | requests-kerberos       | 0.12.0     | tensorflow-model-optimization | 0.7.3       |
| pyasn1                   | 0.4.8   | requests-negotiate-sspi | 0.5.3      | termcolor                     | 2.1.0       |
| pyasn1-modules           | 0.2.8   | requests-oauthlib       | 1.3.0      | terminado                     | 0.17.1      |
| pybind11                 | 2.10.4  | requests-toolbelt       | 1.0.0      | terminaltables                | 3.1.0       |
| pybind11_json            | 0.2.13  | rich                    | 13.3.5     | testpath                      | 0.5.0       |
| pycocotools              | 2.0.2   | rise                    | 5.7.1      | textwrap3                     | 0.9.2       |
| pycodestyle              | 2.8.0   | rsa                     | 4.7.2      | tflite-model-maker            | 0.3.4       |
| pycparser                | 2.21    | rtree                   | 0.9.4      | tflite-support                | 0.4.1       |
| pyct                     | 0.5.0   | ruamel.yaml             | 0.17.21    | theano-pymc                   | 1.1.2       |
| pyee                     | 8.2.2   | ruamel.yaml.clib        | 0.2.6      | thinc                         | 7.4.0       |
| pyflakes                 | 2.4.0   | s3transfer              | 0.6.0      | threadpoolctl                 | 2.2.0       |
| pygments                 | 2.14.0  | sacremoses              | 0.0.43     | tiffifile                     | 2023.4.12   |
| pyjwt                    | 2.4.0   | saspy                   | 4.3.2      | timm                          | 0.4.12      |
| pylerc                   | 4.0     | scikit-image            | 0.19.3     | tinycss2                      | 1.1.1       |
| pymc3                    | 3.11.4  | scikit-learn            | 1.0.2      | tokenizers                    | 0.10.1      |
| pyodbc                   | 4.0.34  | scikit-plot             | 0.3.7      | toml                          | 0.10.2      |
| pyopenssl                | 23.2.0  | scipy                   | 1.7.3      | tomli                         | 2.0.1       |
| pyarsing                 | 3.0.9   | seaborn                 | 0.12.2     | toolz                         | 0.12.0      |
| pypdf                    | 3.15.0  | semver                  | 2.13.0     | torch-cluster                 | 1.5.9       |
| pypdf2                   | 1.26.0  | send2trash              | 1.8.0      | torch-geometric               | 1.7.2       |
| pyperclip                | 1.8.2   | sentencepiece           | 0.1.99     | torch-scatter                 | 2.0.7       |
| pyppeteer                | 1.0.2   | setuptools              | 67.7.2     | torch-sparse                  | 0.6.10      |
| pyquaternion             | 0.9.9   | shap                    | 0.39.0     | torch-spline-conv             | 1.2.1       |
| pyrsistent               | 0.18.0  | shapely                 | 1.7.1      | torchvision                   | 0.9.2       |
| pyshp                    | 2.1.3   | simplejson              | 3.17.6     | tornado                       | 6.3.2       |
| pysocks                  | 1.7.1   | six                     | 1.16.0     | tqdm                          | 4.65.0      |
| pyspnego                 | 0.9.2   | slicer                  | 0.0.7      | traitlets                     | 5.5.0       |
| pytables                 | 3.7.0   | snappy                  | 1.1.10     | transformers                  | 4.5.1       |
| pytest                   | 7.2.0   | sniffio                 | 1.2.0      | trimesh                       | 2.35.39     |
| python                   | 3.9.18  | snuggs                  | 1.4.7      | typed-ast                     | 1.4.3       |
| python-certifi-win32     | 1.6     | sortedcontainers        | 2.4.0      | typeguard                     | 2.13.0      |
| python-dateutil          | 2.8.2   | soupsieve               | 2.5        | typing                        | 3.10.0.0    |
| python-fastjsonschema    | 2.16.2  | spacy                   | 2.2.4      | typing-extensions             | 4.7.1       |
| python-flatbuffers       | 1.12    | sparqlwrapper           | 1.8.5      | typing_extensions             | 4.7.1       |
| python-graphviz          | 0.20.1  | sqlalchemy              | 1.4.39     | tzdata                        | 2023c       |
| python-gssapi            | 1.8.2   | sqlite                  | 3.41.2     | ujson                         | 5.4.0       |
| python-sounddevice       | 0.4.4   | srsly                   | 1.0.5      | urllib3                       | 1.26.16     |
| python_abi               | 3.9     | stack_data              | 0.5.1      | vc                            | 14.2        |
| pytorch                  | 1.8.2   | statsmodels             | 0.14.0     | vs2015_runtime                | 14.27.29016 |
| pytz                     | 2022.6  | stevedore               | 3.3.0      | wasabi                        | 0.9.1       |
| pyviz_comms              | 2.3.0   | swat                    | 1.13.2     | wcwidth                       | 0.2.5       |
| pywavelets               | 1.4.1   | sympy                   | 1.9        | webencodings                  | 0.5.1       |
|                          |         | tabulate                | 0.8.9      |                               |             |

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| Library Name       | Version  |
|--------------------|----------|
| websocket-client   | 0.58.0   |
| websockets         | 10.4     |
| werkzeug           | 2.2.3    |
| wheel              | 0.37.1   |
| widgetsnbextension | 3.6.1    |
| win_inet_pton      | 1.1.0    |
| wincertstore       | 0.2      |
| winkerberos        | 0.8.0    |
| winpty             | 0.4.3    |
| wordcloud          | 1.8.1    |
| wrapt              | 1.14.1   |
| x86cpu             | 0.4      |
| xarray             | 2023.6.0 |
| xeus               | 3.0.5    |
| xeus-python        | 0.15.8   |
| xeus-zmq           | 1.0.2    |
| xgboost            | 1.5.0    |
| xlrd               | 2.0.1    |
| xlwt               | 1.3.0    |
| xmltodict          | 0.12.0   |
| xtl                | 0.7.5    |
| xz                 | 5.4.2    |
| yaml               | 0.2.5    |
| yapf               | 0.31.0   |
| yaml               | 1.8.1    |
| zeromq             | 4.3.4    |
| zfp                | 1.0.0    |
| zict               | 3.0.0    |
| zipp               | 3.11.0   |
| zlib               | 1.2.13   |
| zlib-ng            | 2.0.7    |
| zstd               | 1.5.5    |

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